

# Technical Document #671: Database Systems - Comprehensive Overview

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Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

### Key Takeaways:

- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Database caching stores frequently accessed data in memory.
- Sharding distributes data across multiple database servers.